

Customer Segments

Primary End-Users: Visually impaired individuals (completely blind or low vision) who need assistance with navigation, reading, and object finding.

Elderly Population: Seniors experiencing age-related macular degeneration or general vision loss who want to maintain independence.

B2B & Institutional Clients: Blindness rehabilitation centers, NGOs (like the National Federation of the Blind), and government accessibility programs looking for cost-effective assistive tools to distribute.

Revenue Streams

Hardware Sales: One-time purchase of the Pixelsense device to cover the \$25 production cost.

SaaS/Subscription Model: A low-cost monthly subscription to cover the \$1.50(approx.) per user of server & Gemini API cost, ensuring recurring revenue.

B2B Bulk Licensing: Selling bulk hardware and enterprise API access to institutions and rehabilitation centers.

Channels

Direct-to-Consumer (D2C): An accessible e-commerce website optimized for screen readers.

Institutional Partnerships: Distributing through NGOs, blind advocacy groups, and government disability support networks.

Value Proposition

Advanced AI capabilities: State-of-the-art scene description, text recognition and initial object detection powered by Gemini 2.5 Flash with real-time ViT tracking with minimal ongoing cost (~\$1.5/month).

Modular Architecture: Engineered for easy maintenance, our modular design significantly reduces long-term repair costs and downtime.

Real-time and Offline safety: The 30ms response time of the TOF sensor provides guaranteed offline collision avoidance, ensuring physical safety.

Key Partnerships

AI & Cloud Providers: Google Cloud (for TTS) and Google (for Gemini API access).

Manufacturing Partners: PCB fabricators and electronics suppliers [Espressif Systems (Shanghai) Co. Ltd., InvenSense inc. & Analog Devices inc.] to scale the \$25 BOM efficiently.



Key Resources

Hardware & Firmware: The ESP32-S architecture, camera, audio components, and the firmware that manages UDP streaming and sensor states.

Software Infrastructure: The Python-based server backend, the ViT tracking models, and the integration pipeline with Gemini API and Google Cloud TTS.

Cost Structure

Hardware Production (COGS): \$25 per unit (components, PCB, housing, assembly).

Cloud & API Operational Costs: ~\$1.60 per active user per month (Server hosting, Gemini API calls, Google Cloud TTS).

Customer Acquisition & Marketing: Website maintenance, accessible marketing materials, and outreach programs.

Logistics: Shipping, warehousing, and handling returns.

Key Activities

Hardware Manufacturing: Sourcing generic components and PCB assembly.

Software Maintenance & Optimization: Maintaining server uptime, managing API rate limits, and refining through R&D.